LECTURE 2
STORAGE TYPES AND ITS ANSLYSIS

TYPES OF ENERGY STORAGE USED IN EVS AND HEVS

Several energy storage technologies are used or investigated for EV and HEV applications, including batteries, supercapacitors (SCs), ultracapacitors (UCs), flywheel energy storage systems (FWs) and fuel-cell-based systems (FCs). Batteries provide relatively high specific energy and are suitable for storing the energy required for vehicle propulsion. Supercapacitors and ultracapacitors offer high specific power and rapid charge–discharge capability, making them useful for acceleration assistance and regenerative braking. Flywheel systems store energy as rotational kinetic energy and can provide rapid power delivery. Fuel cells convert the chemical energy of a supplied fuel, such as hydrogen, into electrical energy and can serve as a primary electrical power source in fuel-cell electric vehicles.

Each technology has different characteristics in terms of specific energy, specific power, efficiency, response time, cycle life, cost, safety and maintenance requirements. Consequently, the selection of an appropriate energy storage system depends on the vehicle type, driving cycle, power demand, required driving range and operating conditions. In many applications, a hybrid energy storage system (HESS) combines two or more complementary technologies, such as a battery and a supercapacitor, to achieve a suitable balance between energy capacity and power capability.

Therefore, the study and comparison of energy storage technologies are important for understanding their operating principles, mathematical models, performance characteristics and suitability for EV and HEV applications. Proper selection, sizing and energy management of these systems can improve vehicle efficiency, extend component life, enhance regenerative-energy recovery and support the development of reliable and sustainable transportation systems.

BATTERIES

The batteries are made of unit cells containing the stored chemical energy that can be converted to electrical energy. One or more of these electrochemical cells are connected in series to form one battery. The grouped cells are enclosed in a casing to form a battery module. A battery pack is a collection of these individual battery modules connected in a series and/or parallel combination to deliver the desired voltage and energy to the power electronic drive system

Lead acid batteries that have served the automotive industry over the past 100 years for powering electrical accessories in conventional IC engine vehicles does not have the power and energy density required in electric vehicles and hybrid vehicles. The push for zero-emission vehicles led to numerous research and development efforts on advanced batteries activities in the United States, Europe and Japan. Desirable features sought after in alternative battery technologies are high power and energy density, faster charge time and long cycle life. The research and development progressed slowly until recent years due to the lack of market penetration of electric vehicles. In the meantime, the growth in the electronics industry over the past several decades has led to tremendous advancements in alternative batteries, such as nickel-cadmium (NiCd), nickel-metal-hydride (NiMH) and Li-based batteries technologies. The rechargeable Li-ion battery is the technology of choice for cell phones and laptop computers. Further research on scaling of these battery technologies led to the development of

several battery technologies for electric and hybrid vehicle applications. NiMH battery-packs are currently used in commercially available hybrid electric vehicles, while the Li-ion battery pack is used in the electric vehicle Tesla roadster. Emerging plug-in hybrid vehicles are also likely to use the Li-ion battery technology. While the NiMH and Li-ion batteries are the front runners today for electric and hybrid electric vehicles applications, several other battery technologies have been used in various prototype vehicles. In this section, we will review the promising battery technologies. The future of the battery technologies for electric and hybrid vehicle applications depends on factors including system cost, availability of raw materials, mass production capabilities and lifecycle characteristics. One must note that the electric and hybrid vehicles industry covers a wide spectrum and is not just limited to road vehicles. Some technologies may be more suitable for certain applications for various reasons. Representative properties of the promising batteries technologies along with that of lead acid battery are summarized below in Figure 1 with information obtained from various literatures.

Battery Type	Specific Energy, Wh/kg	Specific Power, W/kg	Energy Efficiency, %	Cycle Life
Lead acid	35–50	150–400	80	500–1,000
Nickel-cadmium	30–50	100–150	75	1,000–2,000
Nickel-metal hydride	60–80	200–400	70	1,000
Aluminum-air	200–300	100	<50	Not available
Zinc-air	100–220	30–80	60	500
Sodium-sulfur	150–240	230	85	1,000
Sodium-nickel chloride	90–120	130–160	80	1,000
Lithium-polymer	150–200	350	Not available	1,000
Lithium ion	90–160	200–350	>90	>1,000

Figure 1: Properties of electric and hybrid electric vehicle batteries

BATTERY CELL STRUCTURE

The energy stored in a battery is the difference in free energy between chemical components in the charged and discharged states. This available chemical energy in a cell is converted into electrical energy only on demand using the basic components of a unit cell; these components are the positive and negative electrodes, the separators and the electrolytes. The electrochemically active ingredient of the positive or negative electrode is called the active material. Chemical reactions take place at the two electrodes, one of which releases electrons, while the other consumes those.

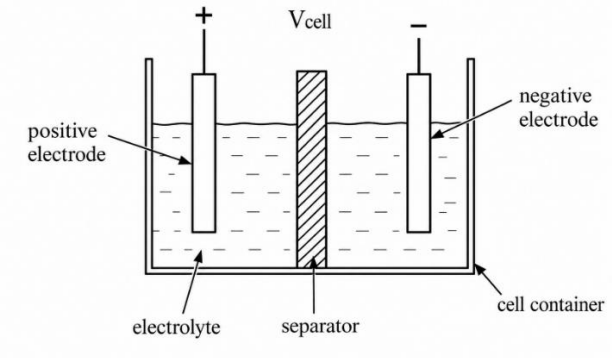


Figure 2: Cell cross-section

The electrodes must be electronically conducting and are located at different sites separated from each other by a separator as shown in Figure 2. The connection points between the electrodes and the external circuit are called the battery terminals. The external circuit ensures

that stored chemical energy is released only on demand and utilized as electrical energy. The components of the battery cell are described in the following:

1. Positive Electrode: The positive electrode is an oxide or sulfide or some other compound that is capable of being reduced during cell discharge. This electrode consumes electrons from the external circuit during cell discharge. Examples of positive electrodes are lead oxide (PbO_2) and nickel oxyhydroxide ($NiOOH$). The electrode materials are in the solid state.
2. Negative Electrode: The negative electrode is a metal or an alloy that is capable of being oxidized during cell discharge. This electrode releases electrons to the external circuit during cell discharge. Examples of negative electrodes are lead (Pb) and cadmium (Cd). The negative electrode materials are in the solid state within the battery cell.
3. Electrolyte: The electrolyte is the medium that permits ionic conduction between positive and negative electrodes of a cell. The electrolyte must have high and selective conductivity for the ions that take part in electrode reactions, but must be a non-conductor for electrons in order to avoid self-discharge of batteries. The electrolyte may be liquid, gel or solid material. Also, the electrolyte can be acidic or alkaline depending on the type of battery. Traditional batteries such as lead acid and NiCd use liquid electrolytes. In lead acid batteries, the electrolyte is the aqueous solution of sulfuric acid ($H_2\ SO_4\ (aq)$). Advanced batteries for electric vehicles, such as sealed lead acid, NiMH and Li-ion batteries use an electrolyte that is gel, paste or resin. Lithium polymer batteries use a solid electrolyte.
4. Separator: The separator is the electrically insulating layer of material, which physically separates electrodes of opposite polarity. Separators must be permeable to the ions of the electrolyte and may also have the function of storing or immobilizing the electrolyte. Present day separators are made from synthetic polymers.

CHEMICAL REACTIONS

During battery operation, chemical reactions at each of the electrodes are sustainable only if electrons generated at the electrodes are able to flow through an external electrical circuit that connects the two electrodes. When a passive electrical circuit element is connected to the electrode terminals of a battery, electrons are released from the negative electrode and consumed at the positive electrode resulting in current flow into the external circuit. In this process, the battery gets discharged. The supply of electrons is due to the chemical reactions at the electrode surfaces inside the battery cell, which are collectively known as reduction and oxidation (redox) reactions. During battery discharge, the positive electrode gets chemically reduced as it absorbs electrons from the external circuit; the negative electrode gets oxidized as it releases electrons to the external circuit. For battery charging, a source with voltage higher than the battery terminal voltage has to be applied so that current can flow into the battery in the opposite direction. During charging, electrons are released at the positive electrode and consumed at the negative electrode; consequently, the positive electrode is oxidized and negative electrode is reduced.

Regardless of the battery cell chemistry, redox reactions take place at the electrodes during both cell charging and discharging for the release and absorption of electrons at the terminals. The generalized redox reactions are given by

$$aA \underset{discharge}{\overset{charge}{\rightleftharpoons}} cC + nE^{+} + ne^{-}$$

for the positive electrode, and

$$bB + nE^{+} + ne^{-} \underset{discharge}{\overset{charge}{\rightleftarrows}} dD$$

for the negative electrode. The combined chemical reaction is

$$aA + bB \underset{discharge}{\overset{charge}{\rightleftarrows}} cC + dD$$

Chemical reactions illustrate that electrons are released and absorbed during any redox reaction. The positive electrode reactions show that during cell charging species A within the electrode is oxidized and becomes energized species C releasing electron(s) into the external circuit and positive ion(s) into the electrolyte. Similarly, the negative electrode reactions show that the species B at the electrode combines with positive ion(s) from the electrolyte and electron(s) from the external circuit to form energized species D. The converse is true at the two electrodes during cell discharging. The coefficients a, b, c and d represent the numbers of moles associated with the species in the reactions; the coefficient n represent the number of electrons and ions involved in the redox reactions.

In electric traction applications, battery cell operation is that of cell discharging when the energy is supplied from the battery to the electric motor for propulsion power and of cell charging when energy is supplied from an external source to store energy in the battery. In conventional vehicles, battery cells supply power to electrical accessories while discharging, and accept energy from an external device to replenish the stored energy during charging. We will next review the redox reactions during cell charging and discharging in a few battery chemistries starting with the lead acid battery cell. Lead acid is still the battery choice for powering electrical accessories in conventional, electric and hybrid electric vehicles. Figure 3 shows the cell discharge operation of a lead acid battery cell into a passive resistive element. The positive electrode made of lead-oxide (PbO_2) is reduced by consuming electrons and ions. The electron supply is through the external circuit which originates at the negative electrode.

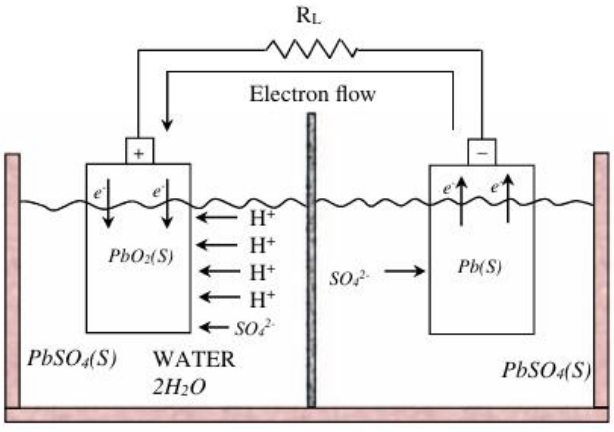


Figure 3:Lead acid battery: cell discharge operation.

The current flow is therefore out of the positive electrode into the electrical load with the battery acting as the source. The positive electrode reaction is given by

$$PbO_2(s) + 4H^{+}(aq) + SO_4^{2-}(aq) + 2e^{-} \rightarrow PbSO_4 + 2H_2O(l)$$

A highly porous structure is used for the positive electrode to increase PbO_2(s)/electrolyte contact area. A porous electrode structure results in higher current densities since PbO2 is converted to $PbSO_4$(s) during cell discharge. As discharge proceeds, the internal resistance of the cell rises due to $PbSO_4$ formation and decreases the electrolyte conductivity as H_2SO_4 is consumed. $PbSO_4$(s) deposited on either electrode in a dense, fine-grain form can lead to sulfation. The discharge reaction is largely inhibited by the buildup of $PbSO_4$, which reduces cell capacity significantly from the theoretical capacity. The negative electrode is made of solid lead (Pb); during cell discharge, lead is oxidized releasing electrons into the external circuit. The negative electrode reaction during cell discharge is

$$Pb(s) + SO_4^{2-}(aq) \rightarrow PbSO_4 + 2e^-$$

Production of $PbSO_4$(s) can degrade battery performance by making the negative electrode more passive.

The overall cell discharge chemical reaction is

$$Pb(s) + PbO_2(s) + 2H_2SO_4(aq) \rightarrow 2PbSO_4 + 2H_2O(l).$$

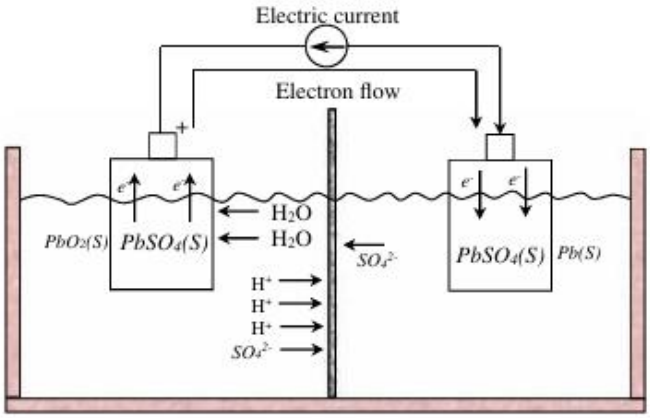


Figure 4: Lead acid battery: Cell charge operation

The cell charge operation is the reverse of the cell discharge operation. An external electrical source supplies current into the battery to reverse the chemical reactions as shown in Figure 4. During cell charging, the lead sulfate is converted back to the reactant states of lead and lead oxide. The electrons are consumed from the external source at the negative electrode, while the positive electrode produces the electrons. The current flows into the positive electrode from the external source thereby delivering electrical energy into the cell where it gets converted into chemical energy. The positive electrode is oxidized releasing electrons during cell charging as follows

$$PbSO_4(s) + 2H_2O(l) \rightarrow PbO_2(s) + 4H^+(aq) + SO_4^{2-}(aq) + 2e^-$$

The negative electrode is reduced during cell charging absorbing electrons, and the chemical reaction is

$$PbSO_4(s) + 2e^- \rightarrow Pb(s) + SO_4^{2-}(aq)$$

The overall chemical reaction during cell charging is

$$2PbSO_4(s) + 2H_2O(l) \rightarrow Pb(s) + PbO_2(s) + 2H_2SO_4(aq).$$

The NiCd and NiMH batteries are examples of alkaline batteries where electrical energy is derived from the chemical reaction of a metal with oxygen in an alkaline electrolyte medium. The specific energy of alkaline batteries is lowered due to the mass addition of the carrier metal. The NiCd battery employs a nickel oxide positive electrode and a metallic cadmium negative electrode. The reactions take place in the presence of potassium hydroxide (KOH) electrolyte. The positive electrode chemical reaction is

$$NiOOH + H_2O + e^- \underset{charge}{\overset{discharge}{\rightleftarrows}} Ni(OH)_2 + OH^-$$

The negative electrode chemical reaction is

$$Cd + 2OH^- \underset{charge}{\overset{discharge}{\rightleftarrows}} cd(OH)_2 + 2e^-$$

The overall chemical reaction is

$$Cd + 2NiOOH + 2H_2O \underset{charge}{\overset{discharge}{\rightleftarrows}} 2Ni(OH)_2 + Cd(OH)_2$$

The reactions at each electrode, which also help to explain 'where the electrons come from' and how the battery works. This battery makes an interesting comparison with the lead acid in that here the electrolyte becomes more concentrated as the cell discharges.

In NiMH batteries, the positive electrode is a nickel oxide similar to that used in a NiCd battery, while the negative electrode has been replaced by a metal hydride which stores hydrogen atoms. The concept of NiMH batteries is based on the fact that fine particles of certain metallic alloys when exposed to hydrogen at certain pressures and temperatures absorb large quantities of the gas to form the metal hydride compounds. Furthermore, the metal hydrides are able to absorb and release hydrogen many times without deterioration. The two electrode chemical reactions in a NiMH battery are as follows:

At the positive electrode,

$$NiOOH + H_2O + e^- \underset{charge}{\overset{discharge}{\rightleftarrows}} Ni(OH)_2 + OH^-$$

At the negative electrode

$$MH_x + OH^- \underset{charge}{\overset{discharge}{\rightleftarrows}} MH_{x-1} + H_2O + e^-$$

M stands for metallic alloy, which takes up hydrogen at ambient temperature to form the metal hydride MH_x ; x is the number of hydrogen atoms absorbed. The overall chemical reaction is

$$NiOOH + MH_x \underset{charge}{\overset{discharge}{\rightleftarrows}} Ni(OH)_2\ MH_{x-1}$$

LEAD ACID BATTERIES

Lead-acid batteries were widely used during the early development of electric vehicles. Introduced in 1859, they are among the oldest rechargeable battery technologies and have a long history of commercial use. Their mature technology, relatively low cost, availability of raw materials, and established manufacturing processes have contributed to their widespread adoption.

Lead-acid batteries offer several advantages, including low cost, ease of manufacturing, high power capability, reliability, and an established recycling infrastructure. Their raw materials are readily available, making them relatively inexpensive compared with many other battery technologies. These characteristics have made them useful in automotive starting systems, industrial vehicles, backup power systems, and some electric-vehicle applications.

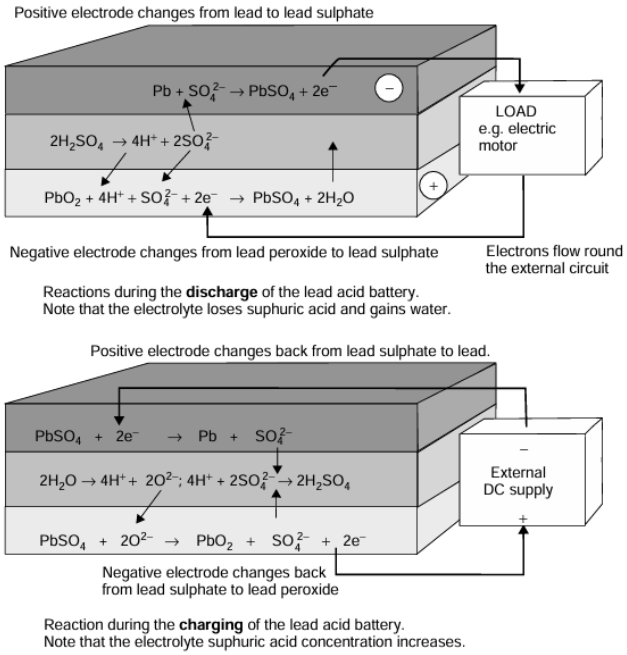


Figure 5: The reactions during the charge and discharge of the lead acid battery

A lead-acid cell has a nominal voltage of approximately 2 V. Therefore, a typical 12 V lead-acid battery contains six cells connected in series. Another important characteristic of lead-acid batteries is their low internal resistance, which enables them to deliver high currents with a relatively small internal voltage drop. This characteristic supports high power delivery, particularly when large currents are required. The capacity of a cell is approximately proportional to the effective plate area, while its internal resistance is approximately inversely proportional to the plate area. Consequently, for a given battery design, internal resistance is approximately inversely proportional to capacity. This is an approximate relationship, as the actual resistance also depends on construction, temperature, state of charge, and other factors.

The internal resistance of a lead-acid battery is important because it affects terminal voltage, power delivery, and efficiency. The empirical relationship used to estimate internal resistance is:

$$R = No.of\ cells \times \frac{0.022}{C_{10}}\ Ohm$$

Special characteristics of lead acid batteries

The normal charging and discharging reactions are not the only reactions that occur in a lead-acid battery. Lead and lead dioxide are not completely stable in sulfuric acid, and slow side reactions can occur at both electrodes. These reactions contribute to self-discharge, even when the battery is disconnected from an external circuit.

At the positive electrode $2PbO_2 + 2H_2SO_4 \rightarrow 2PbSO_4 + 2H_2O + O_2$

and at the negative $Pb + H_2SO_4 \rightarrow PbSO_4 + H_2$

The rate of self-discharge depends on several factors, including temperature, material purity, and the alloys used in the battery. Higher temperatures generally increase the rate of self-discharge.

Cells within a battery may also self-discharge at different rates. Consequently, some cells may become fully charged before others, requiring the charging process to accommodate differences between individual cells. At higher discharge rates, the battery may deliver less usable capacity because of rate-dependent electrochemical behaviour, voltage drop, and other operating effects.

During overcharging, when the energy supplied exceeds that required for the normal charging reactions, electrolysis of water may occur, producing hydrogen at the negative electrode and oxygen at the positive electrode. Gassing occurs primarily near full charge, when less lead sulphate remains available for the normal charging reactions. Under these conditions, additional charging energy can produce hydrogen and oxygen.

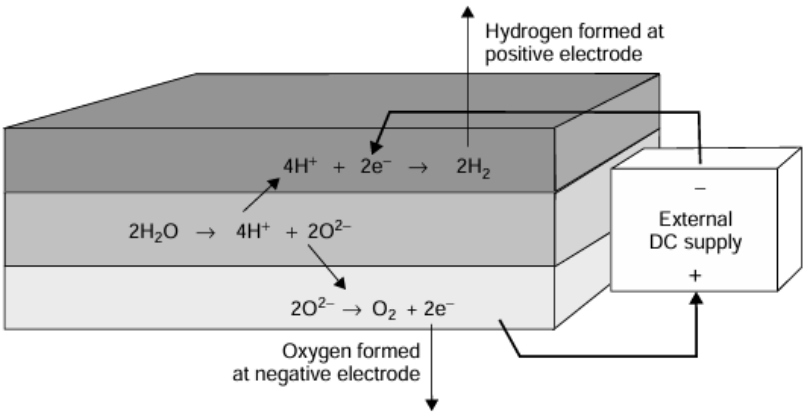


Figure 6: Electrochemical reactions and gas evolution during overcharging of a lead-acid battery.

In conventional flooded lead-acid batteries, the generated gases can escape through the vents, resulting in water loss. Therefore, the electrolyte level must be checked periodically, and distilled water should be added when required and permitted by the manufacturer's instructions.

In sealed lead-acid batteries, oxygen recombination helps reduce water loss. However, excessive charging voltage can increase gas generation and internal pressure. If the pressure exceeds the specified limit, the pressure-relief valve can release gas, causing permanent water loss.

Lead-acid batteries are commonly classified into two broad categories: flooded batteries and valve-regulated lead-acid (VRLA) batteries. Flooded lead-acid batteries contain a free-flowing

liquid sulfuric acid electrolyte that covers the positive and negative plates. The positive plates consist of lead dioxide, while the negative plates consist of sponge lead.

Because the electrolyte is liquid, these batteries generally need to remain upright to minimise leakage. They also require suitable ventilation because hydrogen and oxygen may be produced during charging. Water loss is an important maintenance consideration, and distilled water must be added when required.

VRLA batteries use an oxygen-recombination mechanism to reduce gas loss and minimise electrolyte maintenance. Oxygen generated at the positive electrode travels towards the negative electrode, where it participates in reactions that help reduce water loss.

VRLA batteries incorporate a pressure-relief valve that releases gas when internal pressure exceeds a specified limit. Therefore, they are not completely hermetically sealed, and correct charging conditions remain important.

Two common types of VRLA batteries are gel batteries and absorbent glass mat (AGM) batteries.

Gel batteries: Silica is added to the electrolyte to form a gel, immobilising the electrolyte while allowing gas movement through available pathways.

AGM batteries: The electrolyte is absorbed and retained in a porous glass-fibre mat. The mat immobilises the electrolyte while maintaining pathways for oxygen diffusion.

Both designs reduce the risk of electrolyte leakage and support oxygen recombination.

Feature	**Flooded lead-acid battery**	**VRLA battery**
Electrolyte	Free-flowing liquid	Immobilized in gel or AGM
Risk of leakage	Higher if tilted or damaged	Lower under normal conditions
Gas management	Gas may escape through vents	Oxygen recombination; pressure-relief valve
Free-flowing liquid	May require periodic checking and topping up	Normally maintenance-free regarding water topping-up
Mounting flexibility	Generally, must remain upright	More flexible mounting, subject to manufacturer instructions

Sulfation is a degradation process that can occur when a lead-acid battery remains in a discharged or partially discharged state for an extended period. Lead sulfate forms during normal discharge, but prolonged storage in a discharged condition can cause some of the lead sulfate to develop into larger, more difficult-to-reconvert crystals. This can reduce the battery's ability to accept charge and deliver power. Appropriate recharging may reverse some sulfation, but recovery is not always complete, particularly when the battery has been left discharged for a long time. To reduce the risk of sulfation, lead-acid batteries should be recharged appropriately after discharge and stored according to the manufacturer's recommendations.

Lead-acid batteries require careful handling because they contain corrosive sulfuric acid and toxic lead. Hydrogen generated during charging is flammable and can create an explosion hazard if it accumulates in an inadequately ventilated area. Excessive gas generation can also cause pressure buildup in sealed batteries.

Lead exposure may occur during manufacturing, battery damage, handling, or improper disposal. Suitable personal protection, ventilation, correct charging practices, and responsible recycling are therefore essential for safe operation and environmental protection. Although lead is toxic, the established recycling infrastructure for lead-acid batteries is an important advantage when batteries are collected and processed correctly.

Despite their low cost and high power capability, lead-acid batteries have several limitations for electric-vehicle applications. Their relatively low specific energy means that a larger and heavier battery pack is required to store a given amount of energy. This can increase vehicle mass and limit driving range. Their performance can also deteriorate under cold conditions, and their cycle life may be limited in demanding EV applications, particularly when they experience deep discharges or inappropriate charging.

Cycle-life figures depend on battery design and operating conditions. Your original figures—approximately 800 cycles in certain road-vehicle applications and 1,200–1,500 cycles or around 7–8 years for some industrial electric-truck applications—should be retained only with a suitable reference. The comparison also needs to account for differences in depth of discharge, discharge rate, and operating conditions.

In general, lead-acid batteries remain attractive where low initial cost, high current capability, and recyclability are important. However, their low specific energy, weight, and limitations under demanding cycling conditions restrict their suitability for some modern electric vehicles.

Parameter	Lead-acid	Nickel-cadmium (NiCd)	Nickel-metal hydride (NiMH)	Lithium-ion (Li-ion)
Specific energy (Wh/kg)	20–35	40–55	≈65	90
Energy density (Wh/L)	54–95	70–90	≈150	153
Specific power (W/kg)	≈250*	≈125*	200	300
Nominal cell voltage (V)	2.0	1.2	1.2	3.5
Amp-hour efficiency	≈80%	Good	Quite good	Very good
Internal resistance (Ω/cell, 1 Ah)	≈0.022	≈0.06	≈0.06	Very low
Operating temperature	Ambient	–40 to 80°C	Ambient	Ambient
Self-discharge	≈2%/day	0.5%/day	Up to 5%/day	≈10%/month
Cycle life (to 80% capacity)	Up to 800	≈1200	≈1000	>1000
Recharge time	8 h	1 h	1 h	2–3 h

Figure 7: Comparison of batteries chemistry

NICKEL-CADMIUM BATTERY

The NiCd battery was considered one of the main competitors to the lead-acid battery for use in electric vehicles. NiCd batteries have nearly twice the specific energy of lead-acid batteries, meaning that they can store more energy per unit mass. They have been widely used in various applications, including portable electrical appliances and some electric vehicles. NiCd batteries are available in a variety of sizes and shapes. However, larger battery sizes suitable for electric

vehicles may be difficult to obtain because their traditional market has primarily focused on portable power tools and electronic equipment.

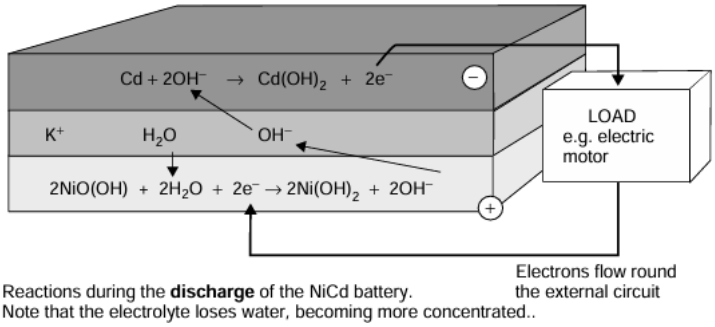


Figure 8: Reactions during discharge of Ni-cd

The operating voltage of a NiCd cell is approximately 1.2 V, while its nominal open-circuit voltage is approximately 1.3 V. Therefore, a larger number of cells must be connected in series to obtain the required battery voltage. For example, a nominal 12 V NiCd battery typically requires 10 cells connected in series, whereas a 12 V lead-acid battery typically requires 6 cells connected in series. This is because the nominal voltage of a NiCd cell is approximately 1.2 V, compared with approximately 2 V for a lead-acid cell. The need for more cells, along with their construction and manufacturing costs, can contribute to the higher cost of a NiCd battery.

The specific energy of NiCd batteries is approximately 40–55 Wh/kg. These batteries offer several advantages, making them suitable for applications requiring high power, reliable operation, and good performance under different environmental conditions.

NiCd batteries have high specific power, allowing them to deliver large amounts of power in a short time. They also offer a long cycle life, reported to be up to 1200 cycles, depending on the battery design and operating conditions. Their wide operating temperature range, from approximately −40°C to +80°C, enables them to operate under both low- and high-temperature conditions.

NiCd batteries have a low self-discharge rate and good long-term storage characteristics. They also exhibit high tolerance to electrical and mechanical abuse, excellent reliability, and low maintenance requirements. Another important characteristic is their relatively flat discharge-voltage profile over a wide range of discharge currents. This means that the battery voltage remains comparatively stable during much of the discharge process.

NiCd batteries also offer rapid charging capability. Some designs can accept approximately 40–80% charge in around 18 minutes under suitable charging conditions. In addition, they generally exhibit better low-temperature performance than lead-acid batteries, making them useful in applications requiring reliable operation at low temperatures.

These characteristics are associated with the stability of the NiCd battery system. Reactions responsible for self-discharge, similar to those occurring in lead-acid batteries, proceed relatively slowly in NiCd batteries. Like lead-acid batteries, NiCd batteries must be charged properly to ensure reliable operation and a long service life. However, because NiCd cells are less prone to self-discharge, the associated charging problems are generally less significant than in lead-acid batteries.

An important feature of NiCd batteries is their ability to tolerate overcharging under specified operating conditions. This is possible because the negative electrode contains an excess of cadmium hydroxide, allowing the positive electrode to become fully charged first. A continuation of the charging current results in the generation of oxygen at the positive electrode via the reaction:

$$4OH^- \rightarrow 2H_2O + O_2 + 4e^-$$

If the charging current continues after the positive electrode is fully charged, oxygen is generated at the positive electrode through an electrochemical reaction. The oxygen then diffuses towards the negative electrode, where it reacts with cadmium to form cadmium hydroxide, using the water produced during the preceding reaction.

$$O_2 + 2Cd + 2H_2O \rightarrow 2Cd(OH)_2$$

Meanwhile, the normal charging reaction continues at the negative electrode, converting cadmium hydroxide back into cadmium. Thus, cadmium hydroxide is produced by the oxygen-recombination reaction and consumed by the normal charging reaction.

$$2Cd(OH)_2 + 4e^- \rightarrow 2Cd + 4OH^-$$

Under suitable operating conditions, the rate of cadmium hydroxide production equals its rate of conversion back into cadmium. Consequently, there is no net consumption of the active materials in the overall process. This mechanism allows NiCd batteries to tolerate a degree of overcharging without a net change in their active materials.

For NiCd batteries, the design allows the overcharging process to continue at a rate of C/10. For example, a 100 Ah battery may be charged at 10 A during this process. Although overcharging results in energy loss, it can be useful during the final stage of charging to equalise the state of charge of individual cells and help bring them to a fully charged condition.

The internal resistance of a NiCd battery is very low; however, it is generally higher than that of a lead-acid battery. This results in a somewhat lower maximum specific power capability compared with lead-acid batteries. The empirical formula for estimating the internal resistance of a NiCd battery is:

$$R = No.of\ cells \times \frac{0.06}{C_3}\ Ohms$$

Comparing this equation with the corresponding empirical equation for lead-acid batteries shows that the coefficient for NiCd batteries is higher: 0.06 compared with 0.022 for lead-acid batteries. In addition, a greater number of cells is required to achieve the same battery voltage because the nominal voltage of a NiCd cell (approximately 1.2 V) is lower than that of a lead-acid cell (approximately 2 V).

The major disadvantages of NiCd batteries are their high cost and the toxicity of cadmium. Cadmium is a toxic heavy metal and a known carcinogen; therefore, proper handling and recycling are essential to minimise environmental contamination and health risks. NiCd batteries may cost approximately three times as much as lead-acid batteries, although their longer service life can partly offset the initial investment. Another limitation is that their charge efficiency can decrease significantly at elevated temperatures, particularly above approximately 35°C, depending on the battery design and charging conditions. Despite their favourable operating characteristics, the high initial cost and environmental concerns

associated with cadmium have restricted the use of NiCd batteries in many electric and hybrid electric vehicle applications.

These limitations encouraged the development and adoption of alternative battery technologies, particularly NiMH batteries. NiMH batteries avoid the use of cadmium and have been widely used in hybrid electric vehicles, making them an important alternative to NiCd technology.

NICKEL-METAL-HYDRIDE (NI-MH) BATTERY

The NiMH battery was introduced commercially in the last decade of the 20th century. Its characteristics are similar to those of the NiCd battery, the main difference being that in the NiMH battery the negative electrode uses hydrogen, absorbed in a metal hydride, which makes it free from cadmium, a considerable advantage. An interesting feature of this battery type is that the negative electrode behaves exactly like a fuel cell. Because of superior specific energy to Ni–Cd and since it is free from toxicity or carcinogenicity, such as cadmium, the Ni–MH battery is superseding the Ni–Cd battery.

The reaction at the positive electrode is the same as for the NiCd cell; the nickel oxyhydroxide becomes nickel hydroxide during discharge. When the battery is discharged, metal hydride in the negative electrode is oxidized to form metal alloy. At the negative electrode hydrogen is released from the metal to which it was temporarily attached, and reacts, producing water and electrons. The reactions at each electrode are shown in Figure 9.

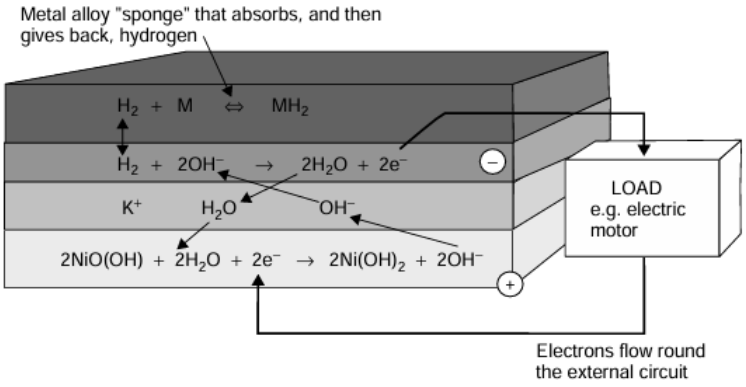


Figure 9: Reaction during discharge of Ni-MH cell

The metals that are used to hold the hydrogen are alloys, whose formulation is usually proprietary. The principle of their operation is exactly the same as in the metal hydride hydrogen stores used in conjunction with fuel cells.

The basic principle is a reversible reaction in which hydrogen is bonded to the metal, and then released as free hydrogen when required. For this to work the cell must be sealed, as an important driver in the absorption/desorption process is the pressure of the hydrogen gas, which is maintained at a fairly constant value. A further important point about the sealing is that the hydrogen-absorbing alloys will be damaged if air is allowed into the cell. This is because they will react with the air, and other molecules will occupy the sites used to store the hydrogen.

NiMH batteries offer improved energy density and, in many designs, higher specific energy and specific power than NiCd batteries. Their nominal cell voltage is approximately 1.2 V, which is the same as that of NiCd cells.

NiMH batteries have a typical specific energy of approximately 60–80 Wh/kg. Their energy density is reported as approximately 150 Wh/L, and their specific power can reach around 200–300 W/kg, depending on battery design and operating conditions. These characteristics make NiMH batteries suitable for applications requiring a balance between energy storage and power delivery.

The main advantages of NiMH batteries include their relatively high specific energy, good specific power, cadmium-free chemistry, relatively flat discharge-voltage profile, and rapid-charging capability under suitable conditions. Their specific energy is generally higher than that of NiCd batteries, allowing more energy to be stored per unit mass. The relatively flat discharge profile also helps maintain a comparatively stable terminal voltage during much of the discharge process.

An important characteristic of NiMH batteries is that the electrolyte composition remains approximately unchanged during normal charging and discharging. Water and hydroxide ions are consumed and regenerated through the electrode reactions, so the electrolyte acts primarily as an ion-conducting medium rather than as a net reactant. This contributes to relatively stable electrolyte composition during normal operation.

The internal resistance and open-circuit voltage of NiMH cells can vary with temperature, state of charge, and other operating conditions. Although NiMH cells can exhibit relatively stable discharge-voltage characteristics, this does not mean their internal resistance or open-circuit voltage remains constant under all conditions. Their internal resistance is of the same general order as that of NiCd cells, although the exact comparison depends on cell design and construction.

Overall, NiMH batteries offer a useful combination of specific energy, power capability, and environmental advantages over NiCd batteries because they do not contain cadmium. These characteristics have supported their use in hybrid electric vehicles and other applications requiring rechargeable energy storage.

One advantage of NiMH batteries over NiCd batteries is their ability to support relatively rapid charging. However, fast charging generates considerable heat. In addition to the heat produced by the battery's internal resistance, the reaction involving hydrogen absorption by the metal alloy at the negative electrode is exothermic, contributing to the overall heat generation during charging. Therefore, effective thermal management is important in NiMH battery systems, particularly in larger battery packs used in electric vehicles. Depending on the battery design and charging rate, cooling fans or other cooling arrangements may be incorporated into the battery system. Small batteries used in electric bicycles and scooters may have simpler cooling requirements, whereas larger battery packs may require more effective cooling systems.

The charging procedure for NiMH batteries is similar in some respects to that of NiCd batteries. The charging current may be controlled or interrupted to keep the cell within appropriate operating limits. NiMH batteries can support charging in approximately one hour under suitable conditions, although the actual charging time depends on cell design, temperature, charging current, and the charging control method.

NiMH cells can also tolerate a limited overcharge current, reported in the source as approximately 0.1C for many cells. Here, C represents the rated battery capacity; for example, 0.1C corresponds to 10 A for a 100 Ah battery. Controlled overcharging may be used during the final charging stage to help ensure that all cells in a battery pack reach a fully charged state.

However, overcharging must be carefully controlled because excessive heat can damage the cells and shorten battery life.

NiMH batteries generally offer slightly higher energy storage capacity than NiCd batteries, but they are also more expensive. One disadvantage of NiMH batteries is their relatively higher self-discharge rate compared with NiCd batteries. Hydrogen can diffuse from the negative electrode through the electrolyte towards the positive electrode, where it participates in a reaction with the nickel oxyhydroxide. This reaction forms nickel hydroxide and effectively discharges the cell, resulting in a loss of stored energy even when the battery is not connected to an external load.

The reaction can be represented as:

$$H_2 + 2NiOOH \rightarrow 2Ni(OH)_2$$

As hydrogen is consumed at the positive electrode, the stored energy of the battery decreases. Consequently, NiMH batteries can lose their charge more rapidly during storage than NiCd batteries, although the actual self-discharge rate depends on cell design, temperature, and storage conditions.

NiMH batteries are well suited for charge-sustaining HEVs because they can withstand thousands of shallow charge–discharge cycles. They repeatedly store regenerative-braking energy and supply power during acceleration while maintaining SoC within a specified range. Their durability and high power capability make them suitable for HEVs, although their specific energy is less suitable for long-range fully electric vehicles.

NiMH batteries offer a relatively long cycle life compared with lead-acid batteries under suitable operating conditions. Their cost is more favourable for high-power applications such as charge-sustaining HEVs than for EVs requiring very high specific energy. Although their components are recyclable, recycling infrastructure may be limited in some regions. Other limitations include relatively low cell efficiency, reduced charge acceptance at elevated temperatures, high initial cost, and possible memory-like effects. Li-ion batteries present a major challenge to NiMH technology, particularly in applications requiring high specific energy.

LI-ION BATTERY

Lithium has a high electrochemical potential and low atomic mass (6.94), making it suitable for producing a battery with a cell voltage of about 3 V when combined with an appropriate positive electrode. The interest in secondary lithium cells soared soon after the advent of lithium primary cells in the 1970s, but the major difficulty was the highly reactive nature of the lithium metal with moisture that restricted the use of liquid electrolytes. The discovery in the late 1970s by researchers at Oxford University that lithium can be intercalated (absorbed) into the crystal lattice of cobalt or nickel to form $LiCoO_2$ or $LiNiO_2$ paved the way toward the development of Li-ion batteries. Since the late 1980s rechargeable lithium cells have come onto the market. They offer greatly increased energy density in comparison with other rechargeable batteries, though at greatly increased cost. Therefore, expensive laptop computers and mobile phones commonly use rechargeable lithium batteries instead of lower-cost NiCd or NiMH cells.

The use of metallic-Li is bypassed in Li-ion batteries by using lithium intercalated (absorbed) carbons (Li_xC) in the form of graphite or coke as the negative electrode along with the lithium

metallic oxides as the positive electrode. The graphite is capable of hosting lithium up to a composition of LiC_6 . The majority of the Li-ion batteries use either a layered oxide or iron phosphates of lithium as the positive electrode. Layered positive electrodes of cobalt oxide are expensive, but proved to be the most satisfactory. Nickel oxide, which costs less can also be used, but is structurally more complex. The performance is similar to that of cobalt oxide electrodes. The manganese oxide-based positive electrodes are also used since manganese is cheaper, widely available and less toxic. Alternative positive electrode material is the lithium iron phosphate ($LiFePO_4$) which can deliver stable and good performance at lower costs.

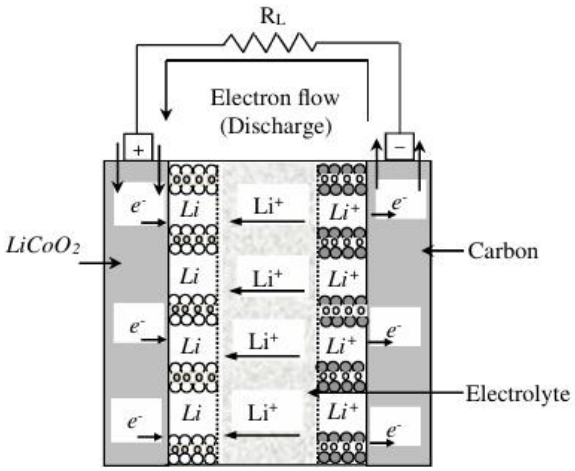


Figure 10: Lithium Ion cell

The cell discharge operation in a Li-ion cell using $LiCoO_2$ is illustrated in Figure 10. During cell discharge, lithium ions (Li+) are released from the negative electrode that travel through an organic electrolyte toward the positive electrode. In the positive electrode, the lithium ions are quickly incorporated into the lithium compound material. The process is completely reversible. The chemical reactions at the electrodes are as follows:

At the negative electrode,

$$Li_xC_6 \underset{charge}{\overset{discharge}{\rightleftarrows}} 6C + xLi^+ + xe^- where\ 0 < x < 1$$

At the positive electrode,

$$xLi^+ + xe^- + Li_{(1-x)}CoO_2 \underset{charge}{\overset{discharge}{\rightleftarrows}} LiCoO_{\cdot 2}$$

During cell charge operation, the lithium ions move in the opposite direction from the positive electrode to the negative electrode

Lithium-ion batteries have a considerable weight advantage over many traditional battery systems, making them attractive for electric vehicles. Their specific energy can be substantially higher than that of lead-acid batteries, potentially enabling a longer driving range for a given battery mass.

The nominal voltage of a typical Li-ion cell is approximately 3.6–3.7 V, compared with about 1.2 V for a typical NiMH or NiCd cell. Thus, one Li-ion cell provides approximately the nominal voltage of three NiMH or NiCd cells connected in series. Automotive Li-ion batteries use different chemistries depending on vehicle requirements, including energy density, cost, safety, power capability and cycle life. NMC (lithium nickel manganese cobalt oxide) and NCA (lithium nickel cobalt aluminium oxide) are used where high energy density is important, including many passenger electric vehicles. LFP (lithium iron phosphate) is widely used in electric vehicles and buses because of its cost advantages, thermal stability and cycle-life characteristics. Its nominal cell voltage is typically around 3.2 V.

Typical values reported for these chemistries vary with cell design and manufacturer. The original figures of 150–200 mAh/g for NMC/NCA and approximately 150 mAh/g for LFP should therefore be treated as approximate material-specific capacity values, not universal specifications for every cell.

COMPARISON BETWEEN VARIOUS LI-ION BATTERY CHEMISTRY

Parameter	Lithium-Cobalt	Lithium-Manganese	Lithium-Nickel Manganese	Lithium-Phosphate	Lithium-Aluminium	Lithium-Titanate	Lithium -Sulphur	Lithium-Air
Nominal Voltage (V)	3.6	3.7	3.6	3.2	3.6	2.4	3	2.91
Cycle life	500-1000	300-700	1000-2000	1000-2000	500	3000-7000	80-110	50-100
Energy Density (Wh/Kg)	150-200	100-150	150-220	90-120	200-260	70-80	350-600	1300
Charge/Discharge rate	1C/1C	1C/10C	1C/2C	1C/25C	1C/1C	5C/10C	5C/5C	NA

An important characteristic of Li-ion batteries is that accurate voltage control is required during charging. If the charging voltage is too high, it can damage the battery and create safety risks. If the charging voltage is too low, the battery may not reach its full state of charge. Therefore, suitable chargers with precise voltage regulation are required.

The main drawbacks of Li-ion batteries are their sensitivity to overcharging and excessive discharge. Overcharging can cause oxidation of the electrolyte solvent at the positive electrode and exothermic decomposition of active electrode materials. These reactions can lead to irreversible cell damage and, under severe conditions, thermal runaway or fire. Excessive discharge can also cause irreversible damage and reduce battery life.

The cost of Li-ion battery cells has decreased considerably, and some Li-ion chemistries can have lower cell costs than comparable NiMH batteries. Certain chemistries also use relatively inexpensive materials. However, lower material costs do not necessarily translate directly into lower battery-pack costs because Li-ion battery packs require a battery management system (BMS) to monitor cell voltage, temperature and current and to help maintain safe operating conditions.

The components of Li-ion batteries can be recycled to recover valuable materials. However, the availability and maturity of recycling infrastructure vary by region, and collection, processing and economic viability remain important challenges.

ULTRACAPACITORS OR SUPERCAPACITORS

An ultracapacitor, also called a supercapacitor, is an energy-storage device that stores electrical energy by separating positive and negative charges. It works on the basic principle of

capacitance, similar to a conventional capacitor, but it can store much more charge because of its large electrode surface area and very small charge-separation distance.

A capacitor stores energy in an electric field. The energy stored in a capacitor is given by

$$E = \frac{1}{2}CV^2$$

where E is the stored energy in joules (J), C is the capacitance in farads (F), and V is the voltage across the capacitor in volts (V).

The capacitors are described in terms of capacitance, which is directly proportional to the dielectric constant of the insulating material and inversely proportional to the space between the two conducting plates. The capacitance of a structure in terms of dimensional parameters is given by

$$C = \frac{\varepsilon A}{d}$$

where ε is the permittivity of the dielectric material, A is the area and d is the separation distance of the charges. Thus, capacitance increases when the plate area increases or the distance between the charges decreases. Ultracapacitors use these principles along with porous carbon electrodes to achieve high capacitance.

The construction of an ultracapacitor is different from that of a conventional capacitor. Instead of using a conventional solid dielectric between two plates, an ultracapacitor uses two porous carbon electrodes, an electrolyte, and a separator. The electrodes are made of activated carbon containing very small particles that provide a large internal surface area. This large surface area allows more electrical charge to be stored. The carbon electrodes are impregnated with a conductive electrolyte containing positive and negative ions.

Double-layer capacitor technology is the major approach to achieving the ultracapacitor concept. The basic principle of a double-layer capacitor is illustrated in Figure 11.

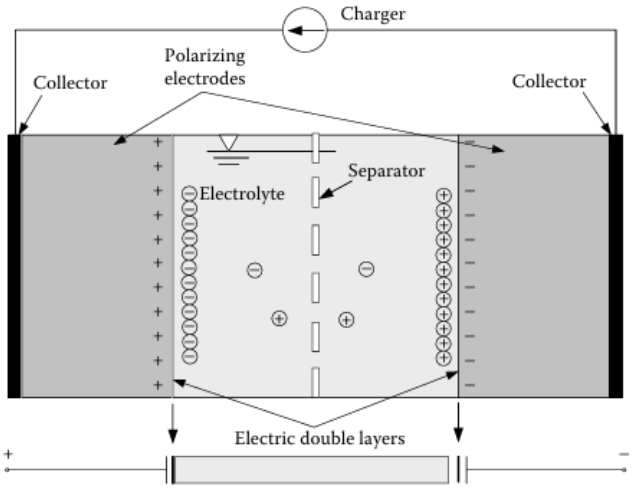


Figure 11: Basic principles of atypical electric double layer capacitor

When two carbon rods are immersed into a thin sulfuric acid solution, separated from each other. A porous separator is placed between the two electrodes. It allows ions to move through the electrolyte but prevents direct electrical contact between the positive and negative electrodes. Metal foils are attached to the carbon electrodes to carry current between the ultracapacitor and the external circuit. The separation between charges is extremely small, approximately 1 nm, which helps the device achieve high capacitance. The operating voltage of an individual ultracapacitor cell is generally low, typically around 2–3 V.

The capacitance of an ultracapacitor can be represented by

$$C = \frac{Q}{V}$$

where Q is the stored charge in coulombs (C) and V is the voltage in volts (V).

Ultracapacitors have high power density, which means they can deliver or absorb a large amount of power in a short time. However, their energy density is generally lower than that of batteries. Conventional capacitors have a power density of approximately $10^{12} W/m^3$ and an energy density of around $50 Wh/m^3$ In comparison, ultracapacitors have a power density of approximately $10^6 W/m^3$ and an energy density of around $10^4 Wh/m^3$. The battery energy density given in the notes is approximately $5 - 25 \times 10^4 Wh/m^3$

Ultracapacitors typically discharge over a period of about 1–10 seconds, whereas batteries can supply energy for much longer periods. Their cycle life can reach approximately (10^5) cycles, compared with around 100–1,000 cycles for batteries in the given comparison. These characteristics make ultracapacitors suitable for applications that require frequent charging and discharging and sudden changes in power demand.

In electric vehicles, ultracapacitors provide high power for short periods and absorb energy quickly. During acceleration, they supply additional power to the electric motor. While climbing a hill, they support the battery by providing extra power when required. During regenerative braking, they absorb and store some of the energy recovered when the vehicle slows down.

Ultracapacitors are generally used along with batteries or fuel cells rather than as the main energy source for an electric vehicle. The battery or fuel cell supplies energy for longer periods, while the ultracapacitor handles sudden power demands and rapid energy recovery. This combination helps meet the changing power requirements of an electric vehicle.

FLYWHEELS

A flywheel is an energy-storage device that stores energy in mechanical form as the kinetic energy of a rotating wheel or disk. When the flywheel rotates, it stores energy, and when its rotational speed decreases, the stored energy can be released and used to perform work.

Flywheels have been used in automobiles for a long time. In internal combustion (IC) engines, a flywheel stores energy and helps smooth out the power delivered by the engine's sudden power pulses. However, the energy stored in these conventional flywheels is relatively small because the vehicle must still be able to accelerate quickly.

Flywheels are also being considered for use in electric vehicles (EVs) and hybrid electric vehicles (HEVs). They can work as an additional power source, replace chemical batteries in some applications, or operate together with batteries. However, improvements in their specific energy, size, weight, and safety are needed before they can be widely used as the primary energy source in electric vehicles.

A flywheel stores energy by rotating at a high speed. A simple example is a flat disk spinning about its axis. The energy stored in the rotating disk is released when the flywheel slows down. In an electrically operated flywheel system, the rotor shaft is connected to a motor/generator. During charging, the motor accelerates the rotor and converts electrical energy into rotational kinetic energy. During discharging, the rotating flywheel drives the generator, which converts the stored kinetic energy back into electrical energy. Power electronics are used to control the electrical power supplied to and obtained from the flywheel.

The flywheel can also be connected mechanically to the vehicle wheels through a gearbox and clutch. In this arrangement, the flywheel supplies energy to the wheels as its rotational speed decreases.

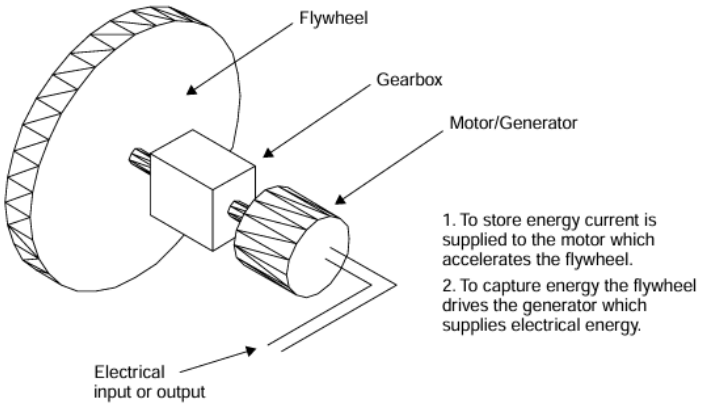


Figure 12: Flywheel/generator arrangement

Flywheels can also be used for regenerative braking. During braking, the kinetic energy of the moving vehicle is used to accelerate the flywheel, storing the vehicle's energy as rotational kinetic energy. This stored energy can later be used to help drive the vehicle.
The energy stored in a rotating flywheel can be expressed as:

$$E = \frac{1}{2} J \omega^2$$

where E is the stored kinetic energy, J is the moment of inertia of the rotating mass, and ω is the angular speed. The equation shows that stored energy increases with the square of rotational speed. Therefore, very high rotational speeds allow a flywheel to store a significant amount of energy without requiring a very large physical mass.
When the flywheel slows down from an initial angular velocity ω_1 to a final angular velocity ω_2, the energy released is

$$\Delta E = \frac{1}{2} I(\omega_1^2 - \omega_2^2)$$

This equation is useful for calculating the amount of energy available from a flywheel during discharge.
The main design objective of a flywheel is to maximize the energy stored per unit mass, known as specific energy. Increasing the rotational speed increases the stored energy, but it also increases the mechanical stress within the flywheel.
The maximum rotational speed is limited by the tensile strength of the material. If the centrifugal stress becomes greater than the material's allowable stress, the flywheel may fail. Therefore, the design must ensure that the stresses remain within safe limits by using an appropriate factor of safety.
The stored energy per unit mass can be expressed as

$$\frac{E}{m} = k \frac{\sigma}{\rho}$$

where: E/m is specific energy (J/kg), k is constant depending on the geometry of the flywheel, σ is Tensile strength of the material (Pa) and ρ is density of the material kg/m^3
This relationship shows that a suitable flywheel material should have high tensile strength and low density. Composite materials, such as carbon-fiber composites, are useful because they can provide high strength with relatively low weight.
Modern composite flywheels can achieve specific energy values of up to approximately $200\ Wh/kg$, as stated in the supplied notes.
Flywheels may be made from steel or composite materials. Steel flywheels have traditionally been used in automobiles, but their energy-storage capacity is limited by their material

properties and weight. Modern flywheels use high-strength composite materials, such as carbon fiber, to increase specific energy and reduce weight. Composite flywheels can also break apart into smaller fragments or a more dispersed mass during a catastrophic failure, unlike steel flywheels, which may produce large, dangerous metal fragments. The choice of material is important because the flywheel must withstand the high stresses produced by rapid rotation while remaining lightweight and safe.

Flywheels have several advantages as energy-storage devices. Their most important advantage is their high specific power, which allows them to supply or absorb large amounts of power in a short time. Theoretical specific power values of approximately $5 - 10\ kW/kg$have been reported, while around $2kW/kg$may be achievable under safe working stresses.

Flywheels are mechanical devices and are not affected by temperature extremes in the same way as chemical batteries. They do not require toxic chemical processing or produce chemical battery waste, making them environmentally attractive in this respect. The energy-conversion efficiency from and to the flywheel can approach 98% , compared with approximately $75 - 80\%$for batteries, according to the supplied notes. Flywheels can also have a long service life, require relatively little maintenance, and recharge quickly. Full recharging in less than 10 minutes may be possible in a suitable flywheel charging station. Their ability to absorb and release high power quickly also makes them useful for regenerative braking.

Despite their advantages, flywheels have several limitations that make their use in electric vehicles challenging. Modern flywheel systems can be complex, large, heavy, and expensive because they require additional equipment for operation, control, and safety. These factors are particularly important in EVs and HEVs, where vehicle weight, space, and cost must be minimized. One major limitation is the loss of energy due to air friction, also called windage loss. To reduce these losses, the flywheel is often enclosed in a vacuum chamber. However, operating in a vacuum creates challenges for the bearings because conventional liquid-lubricated bearings may not be suitable. Magnetic bearings can be used as an alternative, but they add complexity and cost.

Another major concern is safety. A rapidly rotating flywheel contains a large amount of kinetic energy. If it breaks due to excessive stress or damage, the stored energy may be released suddenly, producing dangerous fragments. A strong containment vessel is therefore required to protect the vehicle and passengers in the event of a failure. This containment system adds extra weight and cost. Flywheels also have limited specific energy compared with batteries. Although their specific power is high, they generally cannot store as much energy per unit mass as batteries such as lead-acid or lithium-ion batteries.

FUEL CELL

A fuel cell is an electrochemical device that converts the chemical energy of a fuel directly into electrical energy through a chemical reaction. It works similarly to a battery, but there is an important difference: a battery stores a limited amount of chemical energy internally and requires recharging, whereas a fuel cell can continue producing electricity as long as fuel and an oxidant are supplied.

The basic structure of a fuel cell (see Figure 13) consists of an anode and a cathode similar to a battery. The fuel that is supplied to the cell is hydrogen and oxygen. The concept of fuel cell is the opposite of electrolysis of water where hydrogen and oxygen are combined to form electricity and water. The hydrogen fuel supplied to the fuel cell consists of two hydrogen atoms per molecule chemically bonded together in the form H_2 . This molecule includes two separate nuclei, each containing one proton, while sharing two electrons. The fuel cell breaks apart these hydrogen molecules to produce electricity. The exact nature of accomplishing the task depends

on the fuel cell type, although what remains the same for all fuel cells is that this reaction takes place at the anode. The hydrogen molecule breaks into four parts at the anode due to the chemical reaction releasing hydrogen ions and electrons.

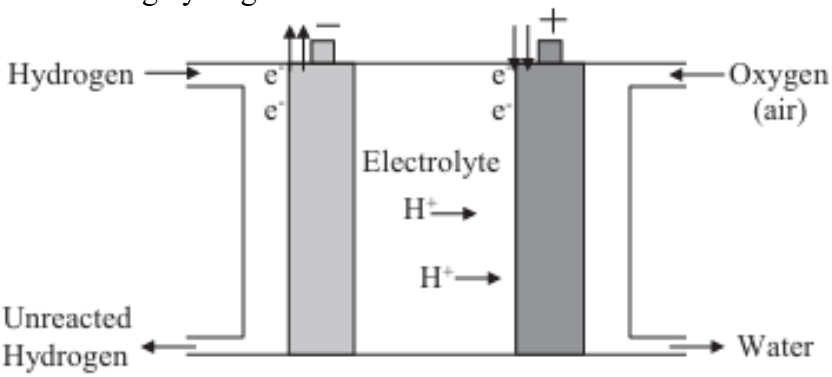


Figure 13: Fuel Cell Structure

A catalyst speeds the reaction and an electrolyte allows the two hydro gen ions, which essentially are two single protons, to move to the cathode through the electrolyte placed between the two electrodes. The flow of electrons from the anode to the cathode through the external circuit is what produces electricity. For the overall cell reaction to complete, oxygen or air must be passed over the cathode. The cathode reaction takes place in two stages. First, the bond between the two oxygen atoms in the molecule breaks and then each ionized oxygen atom grabs two electrons coming from the anode through the external circuit to become negatively charged. The negatively charged oxygen atoms are balanced by the positively charged hydrogen atoms at the cathode and the combination produces H_2O commonly known as water. The chemical reaction tak ing place in a fuel cell is

Anode: $H_2 \rightarrow 2H^+ + 2e^-$

Cathode: $2e^- + 2H^+ + \frac{1}{2}(O_2) \rightarrow H_2O$

Cell: $H_2 + \frac{1}{2}O_2 \rightarrow H_2O$

Fuel cells have several advantages. They can generate electricity continuously as long as fuel and oxygen are supplied. They can offer high energy-conversion efficiency compared with many conventional combustion-based systems. Hydrogen fuel cells produce water as the main reaction product at the point of use, and they generally operate quietly because the fuel-cell stack has few moving parts.

Fuel cells are also suitable for hybrid power systems in which batteries or supercapacitors support rapid changes in power demand. In vehicle applications, hydrogen refuelling can be faster than conventional battery charging when suitable refuelling infrastructure is available.

Despite their advantages, fuel cells face several challenges. Hydrogen production, storage, and transportation can be expensive. Hydrogen must generally be stored at high pressure in fuel-cell vehicles, requiring specialized tanks and safety systems.

Fuel-cell stacks and catalysts can also have high initial costs. Reliable operation requires proper water and temperature management, and the durability of fuel-cell components must be maintained over time. In addition, hydrogen refuelling infrastructure is limited in many regions. The overall environmental impact of a hydrogen fuel-cell system depends on how the hydrogen is produced.

COMPARISON BETWEEN ENERGY STORSGE DEVICE

Parameter	Battery	Ultracapacitor	Flywheel	Fuel cell
Energy storage form	Chemical energy	Electrostatic energy	Mechanical kinetic energy	Hydrogen stores chemical energy; fuel cell converts it into electricity
Working principle	Electrochemical reactions	Separation of electrical charges	Rotation of a rotor at high speed	Electrochemical reaction between fuel and oxidant
Energy density	High	Low	Low to moderate	High at the fuel-system level, depending on hydrogen storage
Power density	Moderate to high	Very high	High	Generally suited to continuous power generation
Charging/refuelling	Requires electrical charging	Very fast charging	Fast acceleration of rotor	Hydrogen refuelling
Response time	Moderate	Very fast	Very fast	Depends on fuel-cell type and system
Cycle life	Limited by battery chemistry and use	Very high	Very high	Limited by stack durability and operating conditions
Main advantage	Stores energy for longer operation	Supplies and absorbs power quickly	Rapid mechanical energy storage and release	Produces electricity continuously while fuel is supplied
Main limitation	Charging time and battery degradation	Low energy capacity	Safety, size, and mechanical complexity	Hydrogen cost, storage, and infrastructure
EV application	Main energy source in battery EVs	Acceleration support and regenerative braking	Short-term energy storage and braking recovery	Main electrical power source in fuel-cell EVs